BCA(I) — Math. Found. (101)

2024

(Session : 2023-26)

(Paper ID : 11001)

Time : 3 hours Full Marks : 80

Note: *Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer **any five** questions.*

1. Find the Eigen values and Eigen vectors of the following matrix:

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

2. Using matrix method, solve the following system of equations:

$$\begin{aligned} x + 2y + 3z &= 6 \\ 2x - y + z &= 5 \\ x + y - 2z &= -2 \end{aligned}$$

3. If $y = (x^{2/3} + x^{-1/3})\ (1 + y^2)\dfrac{dy}{dx} = 2m\dfrac{y}{x^2}$, prove that

$$\frac{1}{m}\,y'' + y' + y + z - m^2 z = 0$$

4. Show that using Maclaurin's series.

$$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$$

5. Evaluate $I = \displaystyle\int \frac{1}{2\cos x - 3\sin x + 5}\,dx$

6. Evaluate $I = \displaystyle\int \frac{\sin x \cos x}{1 + \sin^4 x}\,dx$

7. Evaluate $I = \displaystyle\int_{\pi/3}^{\pi/6} \frac{1}{1 + \sqrt{\tan x}}\,dx$

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

8. Solve

$$\frac{d^2y}{dx^2} + y\tan x = y^2 \sec x$$

9. Solve

$$(x^2 - yx^2)\,dy + (y^2 - xy^2)\,dx = 0$$

10. Write short notes on following:

(a) Inverse matrix

(b) Successive Differentiation

(c) Transpose of matrix

(d) Singular matrix

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